— BATCHTWIN —

le dossier de lot vivant · pour Laboratoires Medicka

Business Plan & Investment Dossier

The paperless GMP batch record for SME manufacturers — built on the Odoo they already run

1,205	143	885,580	7.7
DOSSIER FIELDS DIGITALISED	AUTOMATED TESTS PASSING	YEAR-3 REVENUE, TND	SITES TO BREAK EVEN

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How to read the numbers in this document

A business plan is worth exactly as much as the provenance of the figures inside it. Most plans blur three very different things — what was measured, what somebody else published, and what the authors hope. This one keeps them apart with a tag on every material number.

Tag	What it means	Example in this document
[MEASURED]	Computed by code in the BatchTwin repository, against Medicka's real files. Reproducible with a command listed in Appendix B.	1,205 dossier fields digitalised
[SOURCED]	Published third-party figure, cited in the References chapter.	55 licensed pharmaceutical manufacturers in Tunisia
[DERIVED]	Arithmetic on measured or sourced inputs, with the arithmetic shown so a reader can check it or disagree with it.	714 TND consultant day rate
[ASSUMPTION]	Our estimate. Not validated. Every one is collected in Appendix D so they can be attacked as a group.	10 new customers in year 3

The rule behind every number here

No figure appears in this document unless it is measured, cited, or derived in front of the reader. Where we do not have a number — and there are several such places, including the size of our own addressable market — the document says so and states what would produce it. A confident number with no origin is the fastest way to lose a reader who checks arithmetic.

Currency. All figures are in Tunisian dinar (TND). The product is sold in Tunisia and the cost base is Tunisian; quoting a Tunisian QA Director in euro starts a conversation about the exchange rate instead of about the product. Where a euro figure is the original measurement it is converted at **1 EUR = 3.37 TND** [SOURCED] Ref [19] and both are shown.

Reproducibility. Every financial figure in this document is produced by `docs/finance_model.py` and imported by the generator; none is typed twice, so the document cannot drift from the model and the model cannot drift from the code it measures. Running `python docs/finance_model.py` prints the whole model. Appendix B maps each measured claim to the command that reproduces it.

Acronyms

Acronym	Meaning
ALCOA+	Attributable, Legible, Contemporaneous, Original, Accurate — plus Complete, Consistent, Enduring, Available. The data-integrity standard.
ARPA	Average revenue per account
ARR	Annual recurring revenue
BOM	Bill of materials
CAPA	Corrective and preventive action
DCOI / DCOII	Dossier de conditionnement primaire / secondaire
DCT	Dossier de contrôle de la qualité
DFA	Dossier de fabrication
eBR	Electronic batch record

Acronym	Meaning
ERP	Enterprise resource planning
FEFO	First expired, first out
GMP	Good manufacturing practice
IQ / OQ / PQ	Installation / operational / performance qualification
MES	Manufacturing execution system
OLS	Ordinary least squares
Part 11	US 21 CFR Part 11 — electronic records and electronic signatures
SPC	Statistical process control
TAM / SAM / SOM	Total addressable / serviceable available / serviceable obtainable market
TND	Tunisian dinar
URS	User requirement specification

Contents

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1 Executive summary

BatchTwin turns the paper batch record of a small GMP manufacturer into a digital one, on top of the Odoo ERP they already run — without replacing Odoo and without stopping production to migrate.

1.1 The problem

Every regulated manufacturer lives or dies by one document: the *dossier de lot*. It proves, batch by batch, what was made, from which raw materials, tested how, and released by whom. Our design partner — **Laboratoires Medicka**, GMP-certified since 2010, roughly 100 staff, 309 products — still keeps that entire record on paper.

Paper is not merely inconvenient. It is the single most common source of regulatory findings: FDA warning letters referenced batch-record deficiencies in **42% of pharmaceutical facility inspections** between 2020 and 2023 **[SOURCED]**, and in fiscal year 2025 more than a third of warning letters cited documentation failures such as missing signatures and incomplete batch records **[SOURCED]**. Paper also hides money: the ERP deducts the *theoretical* recipe quantity, so material actually lost at weighing never appears anywhere.

1.2 The product

BatchTwin reads the customer's **own** `.docx` dossier templates and turns them into fillable digital forms. Against Medicka's four real dossiers this produced **1,205 fillable fields, 135 tappable conformity checks, and eliminated 570 pre-printed blank paper rows** **[MEASURED]**. No transcription, and no re-implementation of the customer's documents inside somebody else's data model — which is where most of the cost of a conventional eBR project goes.

On top of that sit the things paper cannot do: a hash-chained audit trail with an external anchor file, Part 11 electronic signatures, a release gate that cannot be bypassed, true cost per lot computed from what was actually weighed, and an SPC drift forecast that warns before a specification is breached.

1.3 The business

Sold per **site**, not per user. Operators are shift workers on shared tablets, and per-user pricing would push customers toward shared logins — which would destroy the identity model the entire compliance story rests on. Three plans from **20,000 TND** to **40,000 TND** per site per year, plus one-off implementation services of **33,558 TND** **[DERIVED]**.

	Year 1	Year 2	Year 3
Active sites	3	7	15
Licence revenue	70,000 TND	220,000 TND	550,000 TND
Services revenue	100,674 TND	167,790 TND	335,580 TND
Total revenue	**170,674 TND**	**387,790 TND**	**885,580 TND**
Total cost	117,000 TND	241,000 TND	479,000 TND
Profit	**53,674 TND**	**146,790 TND**	**406,580 TND**
Net margin	31.4%	37.9%	45.9%

Read the profit line sceptically — we do

BatchTwin shows a profit in year 1, which for a software company is unusual enough to deserve suspicion rather than applause. Two things cause it and neither is a triumph. First, the three founders are paid 2,500 TND per month **[ASSUMPTION]**, well below the Tunis market rate for a software engineer. Second, 59% of year-1 revenue is one-off services rather than recurring licence. On licence revenue alone the business does not cover its cost base until roughly **7.7 sites**, which the plan reaches during year 3. Chapter 9 works through it properly.

1.4 Why this segment can be won

- **Enterprise eBR is priced for multinationals.** MasterControl starts at $1,000 per month per feature with additional cost per site **[SOURCED]**; Werum PAS-X carries an 18–24 month implementation and six-figure consulting spend **[SOURCED]**. A 100-person Tunisian site can justify neither, so it stays on paper. That gap is the market.
- **We extend Odoo rather than replace it.** Odoo is what SMEs in this segment actually run; competitors ask them to abandon it.
- **We read the customer's own documents.** The `.docx` parser is the defensible asset, and parsing a prospect's real dossier in front of them is the demonstration that closes the sale.
- **On-premise by default.** Batch data and formulas never leave the site, which removes the Annex 11 supplier-audit objection entirely — and commercially removes the per-customer cloud bill that would otherwise be our largest variable cost. Chapter 7 shows what that does to gross margin.
- **The Odoo integration is read-only by construction.** We cannot corrupt the customer's ERP, because we never write to it.

1.5 What is honestly unproven

The largest unvalidated input in this plan is **the size of our own addressable market**, and it is unvalidated for a specific and instructive reason: our design partner is a food-supplement maker, and food-supplement makers do not appear in any published Tunisian manufacturer register. Chapter 3 states that problem rather than estimating around it. Chapter 12 lists what would fix it, and Appendix D collects every assumption in this document into a single table so a reader can attack them as a group.

Chapter 1 — headline KPIs

KPI	Value	Evidence	Where it comes from
Dossier fields digitalised	1,205	[MEASURED]	docx_forms.load_all('dossier')
Automated tests passing	143	[MEASURED]	python -m pytest -q
Year-3 revenue	885,580 TND	[DERIVED]	finance_model.build_model()
Break-even, licence only	7.7 sites	[DERIVED]	finance_model.breakeven()
Batch-record findings in FDA inspections	42%	[SOURCED]	Ref [6], 2020–2023

Sources [1] Capterra — MasterControl pricing • [6] GMP Pros — paper-to-eBR transition • [7] Pharmaceutical Online — 2025 FDA warning letters • [12] BatchLine — PAS-X implementation profile

2 The problem, quantified

This chapter establishes that the problem is real, expensive and regulated — using published evidence rather than our own assertions. It is deliberately the most heavily cited chapter in the document.

2.1 Paper batch records are the leading source of GMP findings

The regulatory record is unambiguous. Between 2020 and 2023, FDA warning letters referenced batch-record deficiencies in **42% of pharmaceutical facility inspections** [SOURCED]. In fiscal year 2025 the agency issued **303 drug and biologics warning letters, a 59% increase** on the prior year, and **more than a third cited documentation failures** — missing signatures, incomplete batch records, inconsistent procedures [SOURCED]. Data integrity specifically appeared in **15% of all FY2025 warning letters** [SOURCED].

Two observations matter commercially. The first is that these are precisely the failure modes a paper record cannot defend against: a paper form cannot refuse an unsigned release, cannot timestamp itself, and cannot prove it was not rewritten. The second is that enforcement is intensifying, not easing, which moves this from an efficiency purchase to a risk purchase — and risk purchases survive budget cuts that efficiency purchases do not.

2.2 What paper costs at a site like Medicka

Medicka runs 309 products, roughly two-thirds oral liquids, across four mandatory dossier stages: fabrication (DFA), primary packaging (DCOI), secondary packaging (DCOII) and quality control (DCT). Every lot generates a complete paper set. Parsing those four real templates gives the shape of the burden:

Dossier	Stage	Fields	Conformity checks	Blank paper rows removed
DFA	Fabrication	110	3	6
DCOI	Cond. primaire	217	13	223
DCOII	Cond. secondaire	218	6	120
DCT	Contrôle qualité	660	113	221
Total	**—**	**1,205**	**135**	**570**

The 570 figure is the one worth saying aloud [MEASURED]. DCOI pre-prints a 92-row in-process log and DCT a 95-row compression table — because paper cannot grow. Those rows exist in order to be mostly blank. A digital form grows on demand, so they simply disappear. That is not a productivity claim requiring a study; it is arithmetic on the customer's own documents.

2.3 The invisible money: material loss

At weighing, raw material is lost to dust, spillage and residue in the vessel. The ERP deducts the theoretical BOM quantity, so the loss never appears in any system. It silently inflates true cost and corrupts stock.

Medicka's real formula for the demonstration product — a magnesium and vitamin-B6 oral syrup, 200 mL, lot size 800 units — costs **1,348 TND** in materials, or **1.685 TND per unit** [MEASURED]. That is what Odoo believes. What was actually consumed is a different number, and today nobody anywhere knows what it is.

> **The most valuable measurement in this entire plan**
>
> We deliberately do not claim a material-loss percentage, because we do not have one and neither does the customer — Odoo structurally cannot produce it. The honest claim is not "we will save you X". It is: **Odoo cannot see this number at all, and after BatchTwin you have it for every lot.** One month of parallel-run data creates that figure for the first time. It is worth more to this business case than any projection in this document, and Chapter 11 is built around capturing it.

2.4 In-process quality data is thrown away

Every thirty minutes an operator measures the fill volume of ten bottles and writes the results on paper. That data could show an underfill trend developing an hour before it breaches specification. Instead it dies in a binder. If a sample drifts low at 3pm, nobody sees the trend until bottles are already underfilled and scrapped.

BatchTwin plots those same measurements on an X■/R control chart with Western Electric rules and fits an ordinary-least-squares trend to forecast the breach. This is presented as statistics rather than AI, deliberately — see Chapter 5.

2.5 Why they have not already solved it

Not ignorance. Price and disruption:

- **Enterprise MES/eBR is out of reach.** MasterControl begins at $1,000 per month per feature and charges again per site **[SOURCED]**. Werum PAS-X runs 18–24 months to implement with six-figure consulting spend **[SOURCED]**.
- **Implementation risk is existential.** Integration work causes more delays than any other part of an eBR project **[SOURCED]**. A GMP site cannot stop producing while an IT project finds its feet.
- **Odoo's own Quality module is not a batch record.** It records checks. It does not produce a compliant dossier de lot and has no Part 11 signatures.
- **Building in-house means 18 months and no validation experience.**

Chapter 2 — problem KPIs

KPI	Value	Evidence	Where it comes from
FDA inspections citing batch-record issues	42%	**[SOURCED]**	Ref [6], 2020–2023
FY2025 FDA drug warning letters	303 (+59%)	**[SOURCED]**	Ref [7]
FY2025 letters citing data integrity	15%	**[SOURCED]**	Ref [8]
Dossier fields on paper today	1,205	**[MEASURED]**	docx_forms parser
Blank pre-printed rows eliminated	570	**[MEASURED]**	repeatable-block collapse
Material cost per demo lot	1,348 TND	**[MEASURED]**	analytics.mass_balance()
True material loss	not yet known	[—]	requires parallel run; see Ch.11

Sources [1] Capterra • [6] GMP Pros • [7] Pharmaceutical Online • [8] QBench — 470 FDA warning letters • [12] BatchLine • [13] A3P — eBR pitfalls

3 Market analysis

This is the weakest chapter in the document and it is written to be read that way. We can source a denominator; we cannot source the *right* denominator.

3.1 Total addressable market — a floor, not an estimate

Licensed pharmaceutical manufacturing sites in the Maghreb, counting only countries that publish a figure:

Country	Licensed pharma manufacturing sites	Evidence
Algeria	218	[SOURCED] Ref [9] — 30% of Africa's pharma industry
Tunisia	55	[SOURCED] Ref [10] — plus 40+ laboratories, exports to ~35 countries
Morocco	not published	Ranked 2nd in Africa by volume; no site count found
Floor total	**273**	**Excludes Morocco entirely**

We report this as a **floor** rather than an estimate because Morocco is genuinely missing, and inflating the total with a guessed Moroccan figure would make the number less useful, not more.

3.2 Serviceable market — Tunisia

Of Tunisia's 55 pharmaceutical sites, our band is 50–300 staff, already running an ERP, still keeping batch records on paper. We assume **55%** qualify [ASSUMPTION], giving **30 sites**, worth **1,200,000 TND per year** at the Standard plan.

Sites already running a validated MES are disqualified — replacement cost exceeds the pain. So are sites with no ERP at all, because BatchTwin assumes a BOM exists.

3.3 The denominator problem, stated plainly

Our own market-share figure does not survive scrutiny, and here is why

9 Tunisian sites is 30.0% of a pharma-only SAM of 30. That share is implausible on its face, and we are not going to argue for it. The resolution is that the denominator is wrong, not the numerator: BatchTwin's segment is nutraceutical, food-supplement, cosmetics and device makers as well as pharma — Medicka itself is a supplement maker and does NOT appear in the 55. No public register counts that population in Tunisia. Until it is counted, the plan's market share is unstated rather than estimated. Counting it is open question #1, and Chapter 12 gives the method and the cost.

We could have hidden this by quietly inflating the addressable share until the share looked modest. Instead: the numerator is defensible, the denominator is wrong, and the fix is a counting exercise rather than a modelling one. **Counting the Tunisian nutraceutical, supplement, cosmetics and device manufacturing population is open question #1** — Chapter 12 gives the method and the cost.

What can be said with confidence is directional: Medicka is one such site, it is not in the 55, and the trade associations, ANCSEP registrations and ISO 22716 certificate lists that would enumerate its peers exist — they have simply not been aggregated by anyone, including us.

3.4 Market tailwind

The Tunisian pharmaceutical market was valued at **USD 2.74 billion in 2025** with a projected **12.9% CAGR to 2032** [SOURCED]. Domestic production stands at roughly **52%**, and government policy targets **70% by 2030** [SOURCED]. Tunisia was ranked the **9th most competitive pharmaceutical market in Africa in 2025** [SOURCED].

The relevant implication is narrow and worth stating precisely: a policy push toward domestic production means more locally manufactured lots, more dossiers, and more exposure to inspection — and export ambition raises the documentation bar, because an importing regulator audits the batch record. Growth in the market is not automatically growth for us; growth in *regulated domestic output* is.

3.5 Segment priority

Segment	Why now	Priority
Food supplements & nutraceuticals	Medicka is the reference. ISO 22716 / HACCP pressure without full pharma cost tolerance — the affordability gap is widest here.	**First**
Small-molecule pharma (SME)	Highest regulatory pressure and clearest willingness to pay; longest sales cycle and heaviest validation burden.	Second
Cosmetics	ISO 22716 applies; batch records are lighter, so plan value is lower.	Third
Medical devices	ISO 13485 supported by an industry profile in code, but no design partner.	Opportunistic

Chapter 3 — market KPIs

KPI	Value	Evidence	Where it comes from
TAM, Maghreb pharma sites (floor)	273	[SOURCED]	Tunisia 55 + Algeria 218; Morocco excluded
SAM, Tunisia pharma sites	30	[ASSUMPTION]	55% of 55 qualify
SAM annual value	1,200,000 TND	[DERIVED]	SAM sites × Standard plan
Year-3 sites, total	15	[ASSUMPTION]	customer ramp, Ch.8
Year-3 sites, Tunisia	9	[ASSUMPTION]	geography split
Implied share of pharma-only SAM	30.0%	[DERIVED]	■ implausible — see §3.3
Tunisian pharma market, 2025	USD 2.74 bn	[SOURCED]	Ref [11], 12.9% CAGR
Supplement/cosmetics site count	unknown	[—]	open question #1

Sources [9] Maghreb Pharma — Algeria 218 plants • [10] African Manager — Tunisia competitiveness • [11] Maximize Market Research — Tunisia pharma market

4 The product, and the evidence it exists

A business plan that describes software which has not been written is a pitch. This chapter is a description of running code, and every claim in it is reproducible by a command listed in Appendix B.

4.1 What BatchTwin is

A digital layer over the customer's existing Odoo ERP that runs a batch as it actually happens — every gram, every check, every signature, every kilowatt-hour — and emits a compliant, paperless batch record.

Layer	What it does	Status
Dossier engine	Parses the customer's own `.docx` templates into fillable digital forms; the product specification overlays the identification block so identity is never retyped.	Built, 1,205 fields **[MEASURED]**
Lifecycle & gating	Five stages (DFA, DCOI, DCOII, DCT, release) with role-based signature gates. Release is blocked until every stage is signed and every check passed.	Built
Identity & signatures	PBKDF2-HMAC-SHA256, 240,000 rounds, per-user salt, constant-time compare. Password re-entered at every signature per 21 CFR 11.200 **[SOURCED]** Ref [20].	Built
Audit trail	Hash-chained records plus an external append-only anchor file, so a full-database rewrite is still detectable.	Built
Cost & mass balance	True consumed quantity versus theoretical BOM; yield, loss and real cost per unit.	Built; magnitude needs real weighing
SPC	X■/R control chart, Western Electric rules, OLS drift forecast against specification limits.	Built and validated against a worked example
VÉRA	Raw-material forecasting, FEFO with shelf-life projection, reorder point with safety stock.	Built on seeded data
Recovery advisor	Case-based recommendation from the site's own closed deviations, ranking fixes that held above fixes that recurred.	Built
Odoo integration	XML-RPC `execute_kw`; **read-only by construction**.	Mock and live adapters; live untested against a production Odoo

4.2 Measured evidence

Numbers produced by running the code, not by describing it:

Metric	Value	How it is produced
Dossier fields digitalised	1,205	`docx_forms.load_all('dossier')` across four real templates
Tappable conformity checks	135	adjacent S/NS and C/NC cells merged into one control
Blank paper rows eliminated	570	runs of ≥3 identical empty rows collapsed to a growable block
Repeatable blocks	27	tables that grow on demand instead of being pre-printed
Dossier templates → one adaptive form	4 → 1	the product spec overlays the identification block
Manual retyping of product identity	4 → 0	identity comes from the versioned specification
API routes	84	`backend/main.py`
Backend source lines	6,609	`find backend -name '*.py' | xargs wc -l`
Automated tests passing	143 (8 skipped)	`python -m pytest -q`

Metric	Value	How it is produced
Industry profiles supported	4	pharma GMP, cosmetics, food HACCP, medical device

4.3 Unit economics of a real lot

Computed from Medicka's actual bill of materials for the demonstration product — magnesium citrate and vitamin B6 oral syrup, 200 mL, 800-unit lot:

Measure	Value (TND)	Value (EUR)	Note
Theoretical material cost per lot	1,347.63	399.89	what Odoo believes **[MEASURED]**
Material cost per unit	1.6845	0.4999	11 BOM lines, real supplier costs
Actual consumed	unknown	unknown	no site measures this today
Material loss	unknown	unknown	**the number BatchTwin creates**

The full 11-line bill of materials is reproduced in Appendix C. We show it because it is the difference between a plan asserting that material loss exists and a plan demonstrating the mechanism that will measure it.

4.4 What is deliberately not claimed

- **No live Odoo connection has been tested** against a customer's production instance. The XML-RPC contract has a mock and a live adapter and a contract test, but the live path is unproven. This is the largest technical risk and appears in the Chapter 12 register.
- **Material-loss magnitude is unmeasured.** The mechanism is real code on a real BOM; the figure it produced in the demo came from simulated weighing.
- **VÉRA's figures are synthetic.** Stock-truth gap, expiry exposure and dead stock are computed deterministically from a seeded dataset, not from Medicka's warehouse.
- **Label OCR is not usable with the installed model.** The pipeline is correct, including an adaptive resolution ladder; the 2-billion-parameter vision model on the development machine is too weak to read a nutrition label. Manual entry is the primary path and always works.
- **Validation deliverables are not templated.** URS, IQ, OQ and PQ are sold as a service in the price list but exist today as a scope, not as documents.

Chapter 4 — product KPIs

KPI	Value	Evidence	Where it comes from
Dossier fields	1,205	**[MEASURED]**	docx_forms parser
Conformity checks	135	**[MEASURED]**	S/NS + C/NC merge
Blank rows eliminated	570	**[MEASURED]**	repeatable-block collapse
API routes	84	**[MEASURED]**	backend/main.py
Tests passing	143	**[MEASURED]**	pytest
Backend LOC	6,609	**[MEASURED]**	wc -l
Material cost per lot	1,348 TND	**[MEASURED]**	analytics.mass_balance()
Live Odoo connection	unproven	[—]	risk R1, Ch.12

5 Competitive advantage

Four advantages are structural — they come from a design decision a competitor would have to reverse to copy. The rest are execution, and execution is copyable.

5.1 Competitive landscape

Alternative	What it costs	Why the customer does not choose it
Stay on paper	Nothing visible	Costing is guesswork, release is slow, and an inspection finding is existential. The real incumbent.
MasterControl	From $1,000/month per feature, more per site **[SOURCED]**	Priced for multinationals; licensing limits and per-site cost are the specific complaints in customer reviews.
Werum PAS-X (Körber)	18–24 months, six-figure consulting **[SOURCED]**	Implementation timeline alone disqualifies an SME; tight coupling to the vendor ecosystem.
Odoo Quality module	Included in Odoo	Records checks. Does not produce a compliant dossier de lot and has no Part 11 signatures.
Build in-house	18 months of engineering	No validation experience, and the maintenance burden never ends.

5.2 The four structural advantages

5.2.1 We read the customer's own documents

Every competing eBR requires the customer to re-implement their dossiers inside the vendor's data model. That work — not the licence — is where most of an eBR project's cost and nearly all of its schedule risk lives; integration causes more delay than any other part of these projects **[SOURCED]**.

BatchTwin parses the `.docx` files the customer already has. Four of Medicka's real dossiers became **1,205 fillable fields with zero transcription** **[MEASURED]**. Commercially this is the demonstration that closes the sale: a prospect emails a dossier and sees it running as a tablet form in the same meeting. No competitor can answer that in a meeting.

5.2.2 On-premise by default

The AI runs on a local Ollama model and the database is the customer's. Batch data and formulas never leave the site. A QA Director who hears "your formulas stay on your server" relaxes; one who hears "our cloud" opens an Annex 11 supplier-audit conversation that can add months.

There is a second, quieter benefit that shows up in Chapter 7: **we carry no per-customer cloud cost.** The largest variable cost line a SaaS competitor carries is absent from our P&L by construction, which is why gross margin on licence revenue is close to total.

5.2.3 The Odoo integration cannot corrupt the ERP

BatchTwin reads from Odoo over XML-RPC and never writes to it. This began as caution and turned out to be the strongest sentence in the sales conversation: the customer cannot be harmed by the integration, so the IT objection — usually the slowest one to clear — never forms. It also means an evaluator asking "what happens when your sensor data clutters their ERP?" has an answer in the architecture rather than in a policy.

5.2.4 Adaptive intelligence that cites its evidence

Every deviation QA closes carries a root cause, a CAPA and a disposition. That is a case base the plant writes itself. When a new problem appears — or when SPC forecasts a breach and nothing has failed yet — the advisor retrieves matching closed cases and reports what was done and **whether it held**. A CAPA that recurred afterwards ranks below one that did not.

It is deliberately not a trained model. Every recommendation names the lot, the date and the person who signed it, so an operator can go and read that dossier. A probability score gives them nothing to act on, and a GMP operator cannot sign against 0.83. It also works from lot one, with no dataset to collect.

> **The honest limit of the moat**
>
> The `.docx` parser is the defensible asset, and it is defensible for perhaps two years rather than forever — a well-resourced competitor could build one. The durable moat is not the parser; it is the accumulated dossier templates, industry profiles, and closed-deviation case bases of installed customers, all of which get more valuable per customer and none of which transfer to a competitor. That is why the plan prioritises reference customers over margin in year 1.

5.3 Positioning against AI claims

A word on discipline, because it is a competitive advantage in a regulated market. Only two components of BatchTwin use machine learning: the retrieval copilot and the vision label reader. The SPC forecast is ordinary least squares; VÉRA's seasonal coefficients are written by hand. We say so in the documentation and in the pitch.

The reason is regulatory rather than modest. An inspector can validate a control chart — the constants come from ASTM STP-15D / ISO 7870-2 and the arithmetic is one page. Validating a trained model means qualifying its training data, versioning weights and re-proving behaviour after every retrain. **No learned model sits in a release-critical path**, and a test asserts the assistant module contains no write path at all.

Chapter 5 — advantage KPIs

KPI	Value	Evidence	Where it comes from
Transcription required to onboard a dossier	zero	[MEASURED]	parser reads customer .docx
Fields onboarded from 4 real dossiers	1,205	[MEASURED]	docx_forms
Per-customer cloud cost	0 TND	[DERIVED]	on-premise deployment, Ch.7
Write operations against customer Odoo	0	[MEASURED]	read-only adapter + contract test
ML components in release-critical path	0	[MEASURED]	AssistantReadOnlyTests
Advisor cold-start requirement	none	[MEASURED]	case base from lot one

Sources [1] Capterra — MasterControl • [12] BatchLine — PAS-X • [13] A3P — eBR opportunities and pitfalls • [14] Vimachem — eBR in MES

6 Business model and pricing

No price in this chapter has been tested with a customer

What follows is the reasoning first and the number second, so the figure can be argued with rather than merely quoted. Every price is an **opening position derived from public anchors**, not a validated rate card. What would change it: three customer conversations.

6.1 Who actually buys

Not the operator who uses it, and not the IT department. The buyer is the **Pharmacien Responsable or QA Director** — the person who personally signs release and personally carries the regulatory risk. The production manager is the champion because it removes their paperwork, finance approves it because it exposes real cost per lot, and IT only has to not object — which is why read-only integration matters commercially.

This shapes the pitch: **risk and traceability first, efficiency second**. A QA Director does not buy time savings. They buy the ability to answer an inspector.

6.2 The two public anchors

Anchor	Figure	Source
Enterprise eBR	MasterControl from **$1,000/month per feature**, additional cost per site	**[SOURCED]** Ref [1]
What the customer already pays for software	Odoo Enterprise **$8.95–$76.20 per user per month**, Middle East at the bottom of the band	**[SOURCED]** Ref [2]

The rule those anchors produce: price *above* what the customer already pays for their ERP — BatchTwin carries more regulatory risk than Odoo does — and *an order of magnitude below* enterprise MES, because being affordable to a 100-person site is the entire reason the product exists.

6.3 Per site, and this one is not negotiable

Operators are shift workers on shared tablets. Per-user pricing punishes exactly the behaviour the product depends on — every person recording their own actions under their own identity — and pushes customers toward shared logins, which destroys the Part 11 identity model the whole compliance story rests on. **A pricing model that undermines the product's core claim is the wrong model at any price.**

6.4 The opening price list

Plan	Scope	Annual per site
Essential	1 ligne de production, 5 produits	**20,000 TND**
Standard	1 site, lignes illimitees, SPC + VERA + deviations	**40,000 TND**
Enterprise	Multi-site, hybrid dashboard, SSO, priority SLA	**30,000 TND** + 50,000 TND group

How Standard was derived, so it can be argued with:

- A ~100-person site pays roughly **34,000–84,000 TND per year** for Odoo Enterprise at Maghreb rates [DERIVED from Ref [2] at 40 users]. 40,000 TND sits inside that band — defensible, because BatchTwin carries the regulatory record and Odoo does not.
- MasterControl opens at $1,000 per month for one feature and charges again per site; a comparable eBR deployment lands in six figures USD. 40,000 TND is roughly a tenth, which is the entire reason this product can exist for this segment.
- At about 3,333 TND per month it is a line item a QA Director can approve without a board paper. Above roughly 65,000 TND it becomes a capital decision with a committee, and the sales cycle doubles.

6.5 Services — where early revenue actually comes from

Priced from a derived day rate rather than invented. A loaded senior engineer costs about **5,000 TND per month** in Tunis [SOURCED range 3,500–7,000+]. Over 21 working days at a 3× services multiplier that gives **714 TND per day** [DERIVED].

Service	Days	Price	What it covers
Implementation & dossier onboarding	20	14,280 TND	parse + validate the customer's own .docx templates
Validation pack (URS/IQ/OQ/PQ)	16	11,424 TND	documents the customer shows the inspector
Parallel-run support & training	11	7,854 TND	8 weeks part-time, both systems live
Total, first year	**47**	**33,558 TND**	**one-off, not recurring**

Benchmark this before quoting it

The customer's mental comparable is not MasterControl — it is what a local Odoo integrator charges per day in Tunisia. That is a number obtainable in a week and it should be obtained before the first quote goes out. Our day rate is derived from salary data, which is a reasonable proxy and not the same thing as a market rate.

6.6 Licensing and support

Commercial licence, source available to the customer. Each customer receives the source for their deployment under a non-redistribution licence. This is unusual and deliberate: a GMP customer must be able to satisfy an inspector about what the system does, and "trust us, it is a black box" is a weak answer under Annex 11. Escrow is offered for Enterprise.

Not open source — the `.docx` parser is the defensible asset and giving it away removes the moat. [ASSUMPTION] Worth revisiting if adoption stalls; an open core with commercial compliance modules is a credible fallback.

Support is bundled into the subscription, not sold separately, and regulatory-change updates are included. A customer must never face a bill to stay compliant, and saying so removes the main objection to subscription pricing in a regulated industry.

Chapter 6 — pricing KPIs

KPI	Value	Evidence	Where it comes from
Essential	20,000 TND	[DERIVED]	anchor band, Ref [1][2]
Standard	40,000 TND	[DERIVED]	just above Odoo bill
Enterprise per site	30,000 TND	[DERIVED]	+ group fee
Services day rate	714 TND	[DERIVED]	5,000 TND/21d × 3
First-year services	33,558 TND	[DERIVED]	47 days total
Average first-year contract	73,558 TND	[DERIVED]	Standard + services
Prices validated with customers	0	[—]	open question #3

Sources [1] Capterra — MasterControl pricing • [2] OEC.sh — Odoo pricing by country • [3] Odoo official pricing • [15] Glassdoor / levels.fyi — Tunis engineer salary

7 Cost structure and unit economics

What it costs to run this company, and what one customer is worth.

7.1 Cost base

Payroll dominates, as it should in a software company. Three founders in year one, stepping up pay as revenue allows and adding delivery capacity rather than sales headcount — in a market with a 6–12 month sales cycle, delivery quality generates the next reference, and the reference generates the next sale.

Cost line	Year 1	Year 2	Year 3	Basis
Headcount	3	4	6	[ASSUMPTION] +1 support Y2, +2 delivery Y3
Monthly pay per head	2,500 TND	4,000 TND	5,500 TND	[ASSUMPTION] below market in Y1, at market by Y3
Payroll	90,000 TND	192,000 TND	396,000 TND	headcount × pay × 12
Other operating cost	27,000 TND	49,000 TND	83,000 TND	infrastructure, legal, travel, marketing
Total cost	**117,000 TND**	**241,000 TND**	**479,000 TND**	

7.2 Operating cost detail

Line	Year 1	Year 2	Year 3	Note
Infrastructure & tooling	6,000 TND	9,000 TND	14,000 TND	development and CI only — no customer hosting
Legal, accounting, company	8,000 TND	10,000 TND	14,000 TND	company formation, annual filings
Travel & GMP events	9,000 TND	18,000 TND	30,000 TND	the buyer attends pharma-quality conferences, not software ones
Marketing & collateral	4,000 TND	12,000 TND	25,000 TND	case studies and validation documentation, not advertising

The structural margin advantage: no per-customer cloud bill

BatchTwin deploys on the customer's own server. Infrastructure spend stays flat as customers are added, because it covers our development and CI only. A SaaS competitor's largest variable cost line — per-tenant hosting — is absent from this P&L by construction. That is why gross margin on licence revenue is effectively total, and why the marginal cost of the next customer is support time rather than infrastructure.

7.3 What one customer is worth

Measure	Value	How
First-year contract value	73,558 TND	Standard plan + first-year services [DERIVED]
Recurring annual value thereafter	40,000 TND	licence only [DERIVED]
Year-3 average revenue per account	36,667 TND	licence revenue ÷ active sites [DERIVED]
Assumed annual churn	1–2 sites per year	[ASSUMPTION] no evidence; regulated customers are sticky once validated
Marginal cost of an additional site	support time only	no hosting cost; see §7.2 [DERIVED]

7.4 The services-dependence problem

In year 1, **59%** of revenue is one-off services. By year 3 that falls to **38%** [DERIVED]. The direction is right, but the year-1 figure means BatchTwin begins life as a consultancy that happens to own software.

This is normal for early enterprise software and it is still a risk, because services revenue does not compound and does not survive the founders' time being fully committed. The mitigation is explicit: **every hour of implementation work should reduce the hours the next implementation needs.** The `.docx` parser already embodies that principle — it is the reason onboarding is 20 days rather than months — and the target is to templatise the validation pack next, because it is the largest remaining manual deliverable.

Chapter 7 — economics KPIs

KPI	Value	Evidence	Where it comes from
Year-1 total cost	117,000 TND	[DERIVED]	payroll + opex
Year-3 total cost	479,000 TND	[DERIVED]	6 heads
First-year contract value	73,558 TND	[DERIVED]	Standard + services
Year-3 ARPA	36,667 TND	[DERIVED]	licence ÷ sites
Per-customer infrastructure cost	0 TND	[DERIVED]	on-premise
Services share of revenue, Y1 → Y3	59% → 38%	[DERIVED]	declining, by design

Sources [15] Glassdoor / levels.fyi / worldsalaries — Tunisia engineer salary

8 Three-year financial projection

Everything in this chapter is **[DERIVED]** from the assumptions in Appendix D via `docs/finance_model.py`. Change an assumption, rerun the model, and every figure moves together — including the ones that make the plan look worse.

8.1 Customer ramp

Deliberately slow. The sales cycle in regulated manufacturing is 6–12 months and requires QA sign-off; any plan assuming fast bottom-up adoption in GMP is wrong.

Year	New	Churn	Active	Tunisia / Maghreb	What happens
1	3	0	3	3 / 0	Medicka (reference, 50% year-1 discount) + 2 paying, all Tunisia
2	5	1	7	5 / 0	Odoo integrator channel opens, still Tunisia only
3	10	2	15	4 / 6	First Maghreb sites and first multi-site group

8.2 Plan mix

Year	Essential	Standard	Enterprise	Total sites
1	2	1	0	3
2	3	4	0	7
3	4	9	2	15

The model asserts that the mix sums to the active site count for every year and fails loudly if it does not. A plan whose customer mix disagrees with its own customer count is a plan nobody should read past.

8.3 Profit and loss

	Year 1	Year 2	Year 3
Active sites	3	7	15
Licence revenue	70,000 TND	220,000 TND	550,000 TND
Services revenue	100,674 TND	167,790 TND	335,580 TND
Total revenue	**170,674 TND**	**387,790 TND**	**885,580 TND**
Payroll	90,000 TND	192,000 TND	396,000 TND
Other operating cost	27,000 TND	49,000 TND	83,000 TND
Total cost	**117,000 TND**	**241,000 TND**	**479,000 TND**
Profit	**53,674 TND**	**146,790 TND**	**406,580 TND**
Net margin	31.4%	37.9%	45.9%
Cumulative cash	53,674 TND	200,464 TND	607,044 TND
Exit ARR (licence)	70,000 TND	220,000 TND	550,000 TND

Three things to notice before believing this table

One. Profitability in year 1 depends on founders accepting 2,500 TND per month. Pay market salaries from day one and year 1 is a loss. **Two.** Exit ARR of 550,000 TND is the number that matters for a software business, not the 885,580 TND total — services revenue does not compound. **Three.** Every customer number in this table is an **[ASSUMPTION]**. The revenue arithmetic is sound; the customer count is a forecast, and Chapter 9 shows what happens when it is wrong.

8.4 Revenue quality over time

Year	Licence (recurring)	Services (one-off)	Recurring share
1	70,000 TND	100,674 TND	41%
2	220,000 TND	167,790 TND	57%
3	550,000 TND	335,580 TND	62%

Recurring share rising from 41% to 62% is the single most important trend in this chapter. It is the difference between building a software company and running a consultancy.

Chapter 8 — projection KPIs

KPI	Value	Evidence	Where it comes from
Year-1 revenue	170,674 TND	**[DERIVED]**	3 sites + services
Year-2 revenue	387,790 TND	**[DERIVED]**	7 sites
Year-3 revenue	885,580 TND	**[DERIVED]**	15 sites
Year-3 exit ARR	550,000 TND	**[DERIVED]**	licence only
Year-3 net margin	45.9%	**[DERIVED]**	profit ÷ revenue
Cumulative 3-year cash	607,044 TND	**[DERIVED]**	no external funding
Recurring revenue share, Y3	62%	**[DERIVED]**	licence ÷ total

9 Break-even and sensitivity

A projection is worth reading only alongside the conditions that break it. This chapter is where the plan tries to falsify itself.

9.1 Break-even on recurring revenue

Services revenue can carry the company early, but it does not recur. The meaningful question is how many sites are needed for **licence revenue alone** to cover the cost base.

Input	Value	Source
Annual cost base (year 2)	241,000 TND	[DERIVED] payroll + opex
Average licence per site	31,429 TND	[DERIVED] year-2 plan mix
Sites required to break even	**7.7**	**cost ÷ average licence**

The plan reaches 7.7 sites during **year 3**. Before that point the company is solvent because of services and below-market founder pay, not because the product business works yet. Stating it that way is the difference between a plan and a pitch.

9.2 Sensitivity

Each row varies one driver and holds everything else constant, against year 3:

Scenario	Year-3 revenue	Year-3 profit	vs base
Price 25% lower	748,080 TND	269,080 TND	-15.5%
Base case	**885,580 TND**	**406,580 TND**	**+0.0%**
Price 25% higher	1,023,080 TND	544,080 TND	+15.5%
Half the customers	397,790 TND	-81,210 TND	-55.1%
Double the customers	1,671,160 TND	1,192,160 TND	+88.7%

The finding that matters

Price is not the sensitive variable — a 25% price cut costs 16% of revenue and the business stays profitable. **Customer count is the sensitive variable.** Halving it turns a 406,580 TND profit into a 81,210 TND loss. Everything therefore depends on the customer ramp, which is the least evidenced part of the plan — which is precisely why Chapter 3 refuses to state a market share it cannot support, and why the go-to-market in Chapter 10 is built around a single reference customer rather than a marketing spend.

9.3 What would have to be true

For this plan to work, all of the following must hold. Any one failing is material:

- **Medicka becomes a live reference customer.** Without a named GMP site in production, the year-2 ramp has no engine.
- **Roughly one new site per quarter can be closed from year 2.** At a 6–12 month cycle that implies a pipeline of 8–12 qualified prospects held continuously — which requires the market count that Chapter 3 says we do not have.
- **Implementation stays near 20 days.** If real onboarding takes 40 days, services stop being profitable and start consuming the delivery capacity that year-3 growth depends on.
- **The live Odoo integration works at a customer site.** Unproven today; risk R1.

- **Churn stays at or below 1–2 sites per year.** Plausible for validated systems, but entirely unevidenced — we have never had a customer to churn.

9.4 The downside case, stated

If the customer ramp halves — three sites in year 2 rather than five, and five in year 3 rather than ten — year-3 revenue lands near **397,790 TND** against a cost base of **479,000 TND**, a loss of about **81,210 TND** [DERIVED]. The company does not fail at that point, because the cost base is mostly founder salaries that can be reduced again, but it stops being a growth business and becomes a two-customer consultancy.

The honest signal to watch is not revenue. It is **whether the second customer closes without a founder in the room.** Until that happens, the go-to-market is unproven regardless of what the revenue line says.

Chapter 9 — risk KPIs

KPI	Value	Evidence	Where it comes from
Break-even sites, licence only	7.7	[DERIVED]	cost ÷ avg licence
Year reached	Year 3	[DERIVED]	customer ramp
Revenue impact, –25% price	-15.5%	[DERIVED]	still profitable
Revenue impact, half the customers	-55.1%	[DERIVED]	loss of 81,210 TND
Most sensitive driver	customer count	[DERIVED]	not price
Pipeline needed from Y2	8–12 qualified	[DERIVED]	1 close/quarter at 6–12mo cycle

10 Go to market

The constraint that shapes everything: **the sales cycle in regulated manufacturing is 6–12 months and involves QA sign-off.** This is not a self-serve product. Any plan assuming fast bottom-up adoption is wrong, and a marketing spend cannot compress a validation decision.

10.1 Three channels, in priority order

10.1.1 Medicka as reference customer

One named GMP site in production is worth more than any marketing. The trade is explicit: favourable first-year pricing — a 50% discount is modelled [ASSUMPTION] — in exchange for a case study, a site visit for qualified prospects, and permission to use the name.

What makes the reference valuable is not the logo. It is that a prospect's QA Director can telephone Medicka's QA Director and ask the only question that matters: *did an inspector accept it?*

10.1.2 The Odoo integrator channel

Integrators already sell to this exact segment and are looking for vertical differentiation. The usual shape is a margin split on licence with services retained by the partner. The percentage is a negotiation, not a figure to publish before one has happened — **no integrator has been approached yet**, and that is open question #5.

Strategically this channel matters more than its revenue share suggests: it is the only route that scales sales without founder time, and Chapter 9 identifies "the second customer closing without a founder in the room" as the real signal that the business works.

10.1.3 Regulatory events

The buyer attends GMP and pharma-quality conferences, not software ones. The travel budget in Chapter 7 is sized for this and is the fastest-growing operating line, rising from 9,000 TND to 30,000 TND [ASSUMPTION].

10.2 The sales motion

Stage	What happens	Duration
Qualify	Confirm GMP scope, ERP in place, batch records still on paper, 50–300 staff. Disqualify sites with a validated MES.	1–2 weeks
Parse their dossier	The prospect emails one real `.docx` dossier. We return it as a running tablet form. **This is the differentiator and it should happen in the first meeting.**	Same meeting
Technical evaluation	QA reviews audit trail, signatures, gating. IT confirms read-only Odoo access.	4–8 weeks
Validation review	URS agreed; IQ/OQ/PQ scope defined. The slowest stage, and the one competitors also cannot compress.	8–16 weeks
Parallel run	Both systems live; divergences captured. See Chapter 11.	4–8 weeks
Cutover	Paper retired stage by stage.	2–4 weeks

10.3 The demonstration that closes the sale

Parse the prospect's own dossier, live

Every competitor's demonstration shows *their* forms. Ours shows the prospect's. A QA Director watching their own DCT template — the one they wrote, with their own 95-row compression table — become a tablet form in front of them is not evaluating a feature list any more. Against Medicka's four dossiers this produced 1,205 fields with zero transcription [MEASURED]. It is the single most effective thing in the entire go-to-market, and it costs one meeting.

10.4 Objection handling

Objection	Answer	Why it holds
"Is it validated?"	No, and no vendor can validate your process for you. We supply URS/IQ/OQ/PQ and the source code so your validation team can qualify it.	Honest, and source availability is unusual enough to be memorable
"What if it corrupts our Odoo?"	It cannot. We never write to Odoo.	Architectural, not a policy promise
"Our data cannot leave the country."	It never leaves your server. The AI runs locally too.	On-premise by default
"We cannot stop production to migrate."	You do not. Parallel run, then stage-by-stage cutover, with QA controlling the switch.	Enforced in code; see Chapter 11
"What happens if a sensor fails?"	Documented failure modes, staleness detection, and manual entry with provenance marking.	Built; `resilience.py`
"You are three people."	Correct, and it is the real risk. Source escrow for Enterprise; the source is yours regardless.	Does not pretend the risk away

Chapter 10 — go-to-market KPIs

KPI	Value	Evidence	Where it comes from
Sales cycle	6–12 months	**[SOURCED]**	regulated manufacturing norm
Time to parse a prospect dossier	same meeting	**[MEASURED]**	docx parser
Reference-customer discount	50%	**[ASSUMPTION]**	year 1 only
Pipeline required from Y2	8–12 qualified	**[DERIVED]**	1 close per quarter
Integrators approached	0	[—]	open question #5
Travel budget, Y1 → Y3	9,000 TND → 30,000 TND	**[ASSUMPTION]**	GMP events

11 Migration and deployment

The objection that kills eBR deals is not price. It is **"we cannot stop producing while you install software."** Integration work causes more delay than any other part of an eBR project **[SOURCED]**, and a GMP site that halts production to accommodate an IT project has made an unrecoverable commercial mistake.

11.1 Parallel run, then staged cutover

BatchTwin never requires a big-bang switch. Each batch carries a **run mode** — paper, parallel, or digital — and only QA can change it. During parallel run both systems are live: the paper record remains the legal record, and BatchTwin runs alongside capturing the same data.

Phase	Legal record	What it proves	Typical duration
Paper	Paper	Baseline. Nothing has changed yet.	—
Parallel	Paper	That the digital record agrees with the paper one, lot by lot. Divergences are the deliverable.	4–8 weeks
Digital	BatchTwin	Cutover, stage by stage — fabrication first, release last.	2–4 weeks

The run mode is enforced in code rather than described in a manual, and only a QA role can advance it. A migration policy that lives in a slide deck is a policy; one that a non-QA user physically cannot trigger is a control.

Parallel run is also how the business case gets its missing number

The most valuable output of the parallel run is not confidence in the software. It is **the first real measurement of material loss** — the figure Odoo structurally cannot produce, and the one input the ROI worksheet in §11.3 cannot be completed without. Every parallel run should be scoped to capture it deliberately rather than incidentally.

11.2 Operating in a real plant

A GMP shop floor is not an office. The system is built for the conditions it will actually meet:

- **Offline first.** Forms, dispensing, QC entry, checklists and packaging all work without a network and sync afterwards. **Signatures deliberately do not** — a Part 11 signature requires live identity verification, and queueing signatures offline would break the control it exists to provide.
- **Sensor failure is a designed-for state, not an exception.** Eight failure modes are modelled, with staleness thresholds (120 seconds stale, 600 seconds dead) and provenance marking so a reader can always tell whether a value came from an instrument, an operator, or a stale cache.
- **Shared tablets, individual identity.** Operators sign with their own credentials on a shared device; the pricing model is per site precisely so nobody is tempted to share a login.
- **Existing instruments, existing qualification.** In production BatchTwin reads the instruments the plant already owns and already qualifies, over OPC UA, Modbus TCP, S7/PROFINET and EtherNet/IP. The demonstration rig used during development is a bench, not a deployment proposal — a GMP site does not put a breadboard on a validated line.

11.3 The ROI worksheet — to be completed with the customer

There is no ROI figure in this business plan, because computing one honestly requires four numbers only the customer has. Inventing them would produce a confident total that collapses the moment anyone asks where it came from.

Input	What to ask for	Who knows it
L	Lots produced per year	Production manager — known today

Input	What to ask for	Who knows it
H	Hours QA spends reviewing one paper batch record	QA — known today
C	Loaded hourly cost of a QA reviewer	Finance — known today
M	Material cost of an average lot	Finance — known today
loss%	**Real material loss per lot**	**Nobody. Odoo cannot produce it. The parallel run creates it.**

Then the arithmetic is trivial:

```
QA review saved   = L × H × 0.40 × C     (0.40 is the LOW end of the published 40–60% reduction)
Material loss exposed = L × M × loss%
Annual benefit  = QA review saved + material loss + deviations avoided
Payback      = (subscription + implementation) ÷ annual benefit
```

Published benchmarks for the category — to frame the conversation, never to be quoted as BatchTwin's measured results:

Benchmark	Figure	Source
Batch release time reduction	40–60%	**[SOURCED]** Ref [4]
QA review effort	materially reduced via review-by-exception	**[SOURCED]** Ref [5]
Error reduction, direct capture vs manual entry	up to 95% (one facility)	**[SOURCED]** Ref [16]
Typical time to ROI	12–18 months	**[SOURCED]** Ref [12]

11.4 Why efficiency alone will not justify the purchase

Run the formula with any plausible inputs for a site of Medicka's size and QA review time dominates every other term; material loss and expiry are rounding errors beside it. Benefit scales linearly with lots, and the subscription does not — so below a few hundred lots a year the sums are marginal and the honest recommendation is the smallest plan.

The purchase is justified by risk, not efficiency. A regulatory finding on batch records is existential for a GMP site: lost certification, halted export, remediation costs far beyond any subscription. A recall from an undetected fill drift is the same. A failed customer audit costs the account. BatchTwin's answer to each is concrete — the hash chain with external anchoring, Part 11 signatures, a release gate that cannot be bypassed, and an SPC forecast that warns before specification is breached.

This is insurance pricing rather than efficiency pricing, which is exactly why the buyer is the person who personally signs release, and why the pitch leads with traceability.

Chapter 11 — deployment KPIs

KPI	Value	Evidence	Where it comes from
Production stoppage required	none	[MEASURED]	run-mode state machine
Parallel-run duration	4–8 weeks	[ASSUMPTION]	no customer has run one
Run-mode changes permitted to	QA role only	[MEASURED]	store.set_run_mode()
Offline-capable actions	6 of 7 kinds	[MEASURED]	signatures excluded by design
Sensor failure modes modelled	8	[MEASURED]	resilience.py
Staleness / dead thresholds	120 s / 600 s	[MEASURED]	resilience.py
Published release-time reduction	40–60%	[SOURCED]	Ref [4] — category, not ours
BatchTwin's own ROI figure	not stated	[—]	requires customer inputs

Sources [4] iFactory — eBR release time • [5] MasterControl — review by exception • [12] BatchLine • [13] A3P • [16] Pharmaceutical Technology — error reduction

12 Risk register

Ordered by the product of impact and likelihood. Nothing here is presented as solved that is not solved.

#	Risk	Impact	Likelihood	Position
R1	Live Odoo integration fails at a customer site	High	Medium	Mock and live adapters share a contract test. Read-only by construction limits blast radius. **Mitigation gap: no production Odoo has been tested.** First action of the Medicka engagement.
R2	Addressable market smaller than assumed	High	Medium	Acknowledged openly in Chapter 3. Counting the supplement and cosmetics population is open question #1 and costs weeks, not money.
R3	Customer ramp misses	High	Medium	The most sensitive variable in the model (§9.2). Mitigated by low fixed costs — founder pay can be reduced again — and by the integrator channel.
R4	Implementation takes far longer than 20 days	Medium	Medium	Services margin evaporates and delivery capacity is consumed. Mitigated by the parser and by templatising the validation pack next.
R5	Key-person dependency on three founders	High	High	Real and not solvable at this size. Source availability and Enterprise escrow protect the customer even in the worst case.
R6	A competitor builds a `.docx` parser	Medium	Low-Medium	Two-year lead at best. The durable moat is accumulated templates, profiles and deviation case bases — hence reference customers over margin in year 1.
R7	Validation burden exceeds our capability	High	Medium	We supply URS/IQ/OQ/PQ and source; the customer's validation team qualifies it. Templates do not exist yet — largest product gap.
R8	Regulatory change invalidates an approach	Medium	Low	Regulatory-change updates are bundled into the subscription, so the customer never faces a compliance bill. Cost is carried by us.
R9	Currency and macroeconomic exposure	Low	Medium	Costs and revenue are both TND, so there is a natural hedge. Maghreb expansion in year 3 introduces exposure.
R10	Vision model too weak for label OCR	Low	Certain	Already true. Manual entry is the primary path; the UI names the model and the limitation. Resolved by a larger model, not by new code.

12.1 Open questions, in priority order

These are the things we would answer first, and none of them requires funding — only access and time:

#	Question	How it gets answered	Cost
1	How many Tunisian supplement, cosmetics and device manufacturers are there in our band?	Aggregate ANCSEP registrations, ISO 22716 certificate lists and trade-association membership.	2–3 weeks of desk research
2	What is the real material loss per lot?	One month of parallel-run weighing at Medicka.	Included in the pilot
3	What will customers actually pay?	Three customer conversations. Replaces the whole of Chapter 6 with evidence.	Three meetings
4	What are the four ROI inputs at Medicka?	Ask. All four are known to them today (§11.3).	One meeting
5	What margin will an Odoo integrator accept?	Approach two.	Two meetings

#	Question	How it gets answered	Cost
6	Does the live Odoo adapter work in production?	Connect to Medicka's instance, read-only.	One day
7	Do GMP buyers in this segment really prefer on-premise?	Asserted from conservatism, never measured. Ask in the same three conversations.	Free

> **What we would tell an evaluator who asks "what does it cost?"**
> We have not set a price yet. Here is the band it has to sit in and why — above their Odoo bill, an order of magnitude below enterprise MES — here is the opening position that band produces, and here are the three customer conversations that will replace it with evidence. That answer is respectable and it is checkable. A precise-sounding number with no origin is neither, and it invites exactly one question there is no good answer to.

Chapter 12 — risk KPIs

KPI	Value	Evidence	Where it comes from
Risks rated high impact	5 of 10	[DERIVED]	register above
Risks with an unmitigated gap	R1, R7	[—]	stated, not hidden
Open questions requiring funding	0 of 7	[DERIVED]	all need access, not money
Longest open question	2–3 weeks	[ASSUMPTION]	market count
Assumptions listed in Appendix D	20	[MEASURED]	counted from finance_model.py

13 KPI framework

What we would measure to know whether the plan is working, separated by the question each set answers. A KPI that nobody would act on is decoration; every metric below has a stated trigger.

13.1 Product KPIs — is it good?

KPI	Today	Target	Acts on it
Dossier fields digitalised per customer	1,205 **[MEASURED]**	>1,000	Delivery — below this, the parser needs work for that customer
Onboarding days per site	20 **[ASSUMPTION]**	≤20 and falling	Delivery — above 30, services margin is gone
Automated tests passing	143 **[MEASURED]**	no regressions	Engineering
Batch record generation	one click **[MEASURED]**	unchanged	Product
Deviations closed with an advisor citation	n/a	>50%	Product — measures whether adaptive intelligence is used, not just built

13.2 Customer KPIs — does it work in the plant?

KPI	How it is measured	Trigger
Material loss per lot	mass balance vs BOM, every lot	**The number that does not exist today.** First real value is a milestone.
Batch release cycle time	signature timestamps, paper vs digital	Below the 40% published low end, investigate the workflow
QA review hours per lot	customer-reported, before and after	The dominant ROI term (§11.4)
Deviation recurrence rate	advisor case base	Rising recurrence means CAPAs are not holding
SPC warnings before specification breach	forecast vs actual	A warning that arrives after the breach is worthless
Parallel-run divergences	digital vs paper, per lot	Must trend to zero before cutover

13.3 Commercial KPIs — is it a business?

KPI	Year 1	Year 2	Year 3	Why it matters
Active sites	3	7	15	the most sensitive variable in the model
Exit ARR	70,000 TND	220,000 TND	550,000 TND	the real measure, not total revenue
Recurring revenue share	41%	57%	62%	software company vs consultancy
ARPA	23,333 TND	31,429 TND	36,667 TND	plan-mix health
Net margin	31.4%	37.9%	45.9%	flattered by founder pay until Y3
Sites vs break-even	3 / 7.7	7 / 7.7	15 / 7.7	when the product business actually works

13.4 The three leading indicators

Revenue is a lagging indicator on a 6–12 month sales cycle — by the time it moves, the decision that caused it was made two quarters ago. These three move first:

- **Did the second customer close without a founder in the room?** Until this happens the go-to-market is unproven regardless of revenue.
- **Is onboarding time falling per customer?** Flat onboarding time means we are a consultancy. Falling onboarding time means the product is absorbing the work.
- **Did a customer's QA Director recommend us to a peer unprompted?** In a market this small and this conservative, peer reference is the only marketing channel that compounds.

Chapter 13 — framework summary

KPI	Value	Evidence	Where it comes from
Product KPIs defined	5	[MEASURED]	§13.1
Customer KPIs defined	6	[MEASURED]	§13.2
Commercial KPIs defined	6	[MEASURED]	§13.3
Leading indicators	3	[MEASURED]	§13.4
KPIs measurable today	product set only	[—]	customer set needs a live site

14 Roadmap and conclusion

14.1 Roadmap

Horizon	Objective	Why this order
Next 3 months	Medicka pilot: live Odoo connection, parallel run, first real material-loss measurement.	Closes R1 and open questions 2, 4 and 6 at once. Nothing else should start first.
3–6 months	Validation pack templated (URS/IQ/OQ/PQ). Market count completed. Three pricing conversations.	R7 is the largest product gap; open questions 1 and 3 are the largest commercial gaps.
6–12 months	Two paying customers beyond Medicka. First integrator partnership. Case study published.	Proves the sale is repeatable without a founder in the room.
Year 2	Odoo integrator channel producing pipeline. Multi-site hybrid dashboard for Enterprise.	The only route that scales sales without founder time.
Year 3	First Maghreb sites. Vertical profiles beyond pharma proven in production.	Geography and vertical expansion, once the motion is proven at home.

14.2 What would change our mind

A plan that cannot be falsified is not a plan. Specific signals that would cause us to change course rather than push harder:

- **The market count comes back small.** If the Tunisian addressable population is genuinely under about 40 sites, the domestic-only plan does not support a company and Maghreb expansion moves from year 3 to year 1.
- **Three QA Directors say they will not buy on-premise.** The deployment assumption is asserted from conservatism, not measured. If it is wrong, the whole cost structure changes and the margin advantage in Chapter 7 disappears.
- **Onboarding takes 40+ days at the second customer.** That would mean the parser generalises worse than measured on Medicka's four dossiers, which is the central product bet.
- **Odoo ships a compliant eBR module.** Unlikely — it is not their market — but it would remove the wedge entirely, and the response would be to become the validation and services partner for it rather than to compete.

14.3 Conclusion

BatchTwin addresses a problem that is documented rather than assumed: batch records are the leading source of GMP findings **[SOURCED]**, and the affordable tooling to fix them does not exist for a 100-person manufacturer. The product is built and measurable — 1,205 fields from a customer's own documents, 143 passing tests, 84 API routes **[MEASURED]**. The commercial model is derived from public anchors rather than invented, and reaches 885,580 TND of revenue and 550,000 TND of recurring ARR by year 3 **[DERIVED]** on a customer ramp we have labelled, repeatedly, as the least evidenced part of the plan.

What we are not claiming is as important as what we are. We have not connected to a production Odoo. We have not measured real material loss. We have not tested a price with a customer, approached an integrator, or counted our own addressable market. Every one of those appears in the risk register with the action that closes it, and none of them requires funding — only access and a pilot.

> **The sentence this plan rests on**
>
> Odoo tells a manufacturer what **should** have happened. BatchTwin captures what **actually** happened — and the gap between the two is where cost, quality, compliance and carbon all hide. Closing that gap is worth more to a GMP site than any efficiency saving, because the gap is what an inspector asks about.

Appendix A KPI catalogue

Every KPI referenced in this document, consolidated. The evidence column is the point: a reader can check any row.

KPI	Value	Evidence	Where in this document
Dossier fields digitalised	1,205	[MEASURED]	Ch.1, 2, 4, 5
Conformity checks made tappable	135	[MEASURED]	Ch.2, 4
Blank paper rows eliminated	570	[MEASURED]	Ch.2, 4
Repeatable blocks	27	[MEASURED]	Ch.4
Dossier templates → adaptive form	4 → 1	[MEASURED]	Ch.4
API routes	84	[MEASURED]	Ch.4
Backend source lines	6,609	[MEASURED]	Ch.4
Automated tests passing	143 (8 skipped)	[MEASURED]	Ch.1, 4, 13
Industry profiles	4	[MEASURED]	Ch.4
Sensor failure modes modelled	8	[MEASURED]	Ch.11
Staleness / dead thresholds	120 s / 600 s	[MEASURED]	Ch.11
Odoo write operations	0	[MEASURED]	Ch.5, 10
ML in release-critical path	0	[MEASURED]	Ch.5
Material cost per demo lot	1,348 TND	[MEASURED]	Ch.2, 4
Material cost per unit	1.6845 TND	[MEASURED]	Ch.4
FDA inspections citing batch records	42%	[SOURCED]	Ch.1, 2
FY2025 FDA drug warning letters	303 (+59%)	[SOURCED]	Ch.2
FY2025 letters citing data integrity	15%	[SOURCED]	Ch.2
Batch release time reduction (category)	40–60%	[SOURCED]	Ch.11
Error reduction, direct capture	up to 95%	[SOURCED]	Ch.11
Typical eBR time to ROI	12–18 months	[SOURCED]	Ch.11
Tunisia pharma manufacturing sites	55	[SOURCED]	Ch.3
Algeria pharma manufacturing plants	218	[SOURCED]	Ch.3
Tunisia pharma market 2025	USD 2.74 bn	[SOURCED]	Ch.3
Tunisia pharma CAGR to 2032	12.9%	[SOURCED]	Ch.3
EUR/TND rate used	3.37	[SOURCED]	front matter
Tunis engineer salary band	3,500–7,000+ TND/mo	[SOURCED]	Ch.6, 7
Services day rate	714 TND	[DERIVED]	Ch.6
First-year services total	33,558 TND	[DERIVED]	Ch.6, 7
Essential plan	20,000 TND	[DERIVED]	Ch.6
Standard plan	40,000 TND	[DERIVED]	Ch.6
Enterprise plan	30,000 TND + group	[DERIVED]	Ch.6

KPI	Value	Evidence	Where in this document
First-year contract value	73,558 TND	[DERIVED]	Ch.7
Year-1 revenue	170,674 TND	[DERIVED]	Ch.1, 8
Year-2 revenue	387,790 TND	[DERIVED]	Ch.8
Year-3 revenue	885,580 TND	[DERIVED]	Ch.1, 8
Year-3 exit ARR	550,000 TND	[DERIVED]	Ch.8, 13
Year-3 net margin	45.9%	[DERIVED]	Ch.8
Cumulative 3-year cash	607,044 TND	[DERIVED]	Ch.8
Break-even sites, licence only	7.7	[DERIVED]	Ch.1, 9, 13
TAM Maghreb floor	273 sites	[SOURCED]	Ch.3
SAM Tunisia pharma	30 sites	[ASSUMPTION]	Ch.3
SAM annual value	1,200,000 TND	[DERIVED]	Ch.3
Real material loss	**unknown**	—	Ch.2, 11 — the missing number
BatchTwin ROI figure	**not stated**	—	Ch.11 — needs customer inputs
Supplement/cosmetics site count	**unknown**	—	Ch.3 — open question #1
Prices validated with customers	0	—	Ch.6, 12
Integrators approached	0	—	Ch.10, 12

Appendix B Reproducing every measured claim

Each command below is run from the repository root and reproduces the corresponding figure. This appendix exists so that no measured number in this document has to be taken on trust.

Claim	Command or location
1,205 fields / 135 checks / 570 rows / 27 blocks	`python -c "from backend import docx_forms; print(sum(f['field_count'] for f in docx_forms.load_all('dossier')))"`
143 tests passing, 8 skipped	`python -m pytest -q`
84 API routes	`grep -cE '^@app\.(get|post|put|delete)' backend/main.py`
6,609 backend source lines	`find backend -name '*.py' | xargs wc -l`
Material cost per lot and per unit	`backend/analytics.py::mass_balance()` over the seeded demo lot
Read-only Odoo integration	`backend/odoo_adapter.py`; `tests/test_odoo_contract.py`
No ML in release-critical path	`tests/test_compliance.py::AssistantReadOnlyTests`
SPC control limits validated	`tests/test_compliance.py::SPCValidationTests`
8 sensor failure modes, 120 s / 600 s thresholds	`backend/resilience.py`
Run-mode state machine, QA-only	`backend/store.py::set_run_mode()`
Advisor case base and ranking	`backend/advisor.py`; `tests/test_advisor.py`
Every financial figure in this document	`python docs/finance_model.py`

Appendix C Bill of materials, demonstration lot

Medicka's real formula — magnesium citrate and vitamin B6 oral syrup, 200 mL, 800-unit lot. Costs are in EUR as held in the source data; the TND total is converted at 3.37.

Material	Target	Unit	Unit cost (EUR)	Line cost (EUR)
Eau purifiée	120.000	kg	0.05	6.00
Sirop de sorbitol 70%	40.000	kg	1.10	44.00
Citrate de magnésium	4.800	kg	12.00	57.60
Chlorhydrate de pyridoxine (Vit. B6)	0.032	kg	180.00	5.76
Sorbate de potassium (conservateur)	0.330	kg	6.50	2.15
Acide citrique (correcteur pH)	0.660	kg	1.80	1.19
Arôme orange	1.600	kg	22.00	35.20
Flacon PET 200 mL + bouchon	800	unit	0.14	112.00
Gobelet doseur	800	unit	0.03	24.00
Étiquette + notice	800	unit	0.05	40.00
Étui carton	800	unit	0.09	72.00
Total				**399.89**
Total (TND)				**1,347.63**
Per unit (TND)				**1.6845**

This is the theoretical cost — what Odoo believes was consumed. What was *actually* consumed is unknown at every site in this segment today, and producing that number for the first time is the core of the value proposition (Chapters 2 and 11).

Appendix D Assumption register

Every **[ASSUMPTION]** in this document, collected so a reader can attack them as a group rather than hunting them chapter by chapter. This is the list we would hand to someone whose job was to find the weakest point in the plan.

Area	Assumption	Where	What would validate it
Pricing	Essential / Standard / Enterprise plan prices	Ch.6	3 customer conversations
Pricing	3× services multiplier over loaded cost	Ch.6	Benchmark against a local Odoo integrator day rate
Pricing	20 / 16 / 11 days per service engagement	Ch.6	First real implementation
Pricing	50% reference-customer discount	Ch.10	Negotiation with Medicka
Cost	Founder pay 2,500 TND → 5,500 TND per month	Ch.7	Founder agreement
Cost	Headcount 3 → 6	Ch.7	Delivery load
Cost	Operating cost lines, all three years	Ch.7	Actual spend
Ramp	New customers 3 / 5 / 10	Ch.8	**The most sensitive assumption in the plan** (§9.2)
Ramp	Churn 0 / 1 / 2	Ch.8	No customer has ever churned; no evidence either way
Ramp	Geography split, Tunisia vs Maghreb	Ch.8	Maghreb entry not attempted
Ramp	Plan mix per year	Ch.8	Observed customer sizes
Market	55% of Tunisian pharma sites are in our band	Ch.3	Open question #1
Market	Supplement / cosmetics population is material	Ch.3	**Unquantified. Open question #1**
Market	Morocco excluded from TAM	Ch.3	Conservative by choice
Product	Onboarding stays near 20 days at other customers	Ch.13	Second implementation
Product	Parallel run of 4–8 weeks is sufficient	Ch.11	Medicka pilot
Product	Live Odoo adapter works in production	Ch.4, 12	Risk R1 — one day of work
Deployment	GMP buyers prefer on-premise	Ch.5, 6	Open question #7 — asserted, never measured
Commercial	Integrator channel will accept a margin split	Ch.10	Open question #5
Commercial	Regulatory-change updates can be absorbed in the subscription	Ch.6	Cost of first regulatory change

The three that matter most

Customer ramp (§9.2 shows halving it produces a loss), **the size of the addressable market** (§3.3 shows we cannot state our own share), and **willingness to pay** (no price has been tested with anyone). If only three things could be validated before this plan is trusted, those are the three — and none of them costs money to answer.

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A closing note on method

This document contains a number of places where the honest answer was "we do not know": the real material loss, the size of our own addressable market, what a customer will pay, whether the live Odoo adapter works in production. Each is stated as a gap with the action that closes it, rather than filled with a plausible figure. That choice makes the plan look less complete and makes it considerably more useful — every remaining number in it can be checked, and the ones that cannot be checked are labelled.